

Chapter - 1

Introduction

Monitoring water quality is essential for ecosystem health and safety, with modern Water Quality Monitoring Systems (WQMS) utilizing IoT technology for real-time assessment of critical parameters like temperature and electrical conductivity. These systems deploy highly precise sensors placed strategically in water bodies to capture accurate data. Initial processing of this data occurs locally through ESP32 microcontrollers.

In the cloud, sophisticated algorithms analyze the data comprehensively to extract meaningful insights, detect trends, and identify potential anomalies in water quality metrics. This analytical capability not only provides a detailed understanding of current conditions but also supports predictive modeling and proactive management strategies.

WQMS are particularly suited for environments where immediate and accurate monitoring is crucial. Continuous monitoring of temperature and electrical conductivity enables proactive management by promptly detecting deviations and ensuring swift responses to maintain stability.

The integration of IoT technology within WQMS enhances water management practices by facilitating real-time data collection, transmission, and analysis. It promotes efficient water usage, minimizes wastage, and supports sustainable practices.

Moreover, WQMS serve as educational tools, offering stakeholders opportunities to engage in learning about water quality management principles. They support hands-on activities such as practical demonstrations and data analysis exercises.

In conclusion, modern WQMS play a pivotal role in safeguarding water quality, promoting sustainable practices, and enhancing environmental resilience. By leveraging IoT-driven monitoring, advanced analytics, and stakeholder engagement, these systems support effective water resource management, ensure ecosystem health, and contribute to long-term conservation efforts.

Chapter 2

Literature survey

AUTHORS	OBJECTIVE	METHODOLOGY	FINDINGS
Mishra,A.,& Misra,D. (2018)	Explore IoT for efficient agriculture water monitoring.	Utilize IoT sensors for agriculture water quality	IoT sensors optimize agriculture water quality monitoring
Elakkiya,R (2020)	Develop IoT for water quality management.	IoT integrates sensors for water quality management.	IoT enhances water quality management system development.
Singh, A., & Jain, S. (2017)	Examine IoT applications in water quality.	Study IoT sensor tech for water quality.	IoT enhances real-time monitoring, early detection of water quality issues.
Jain, A. (2019)	Develop an IoT-based system for real-time water quality monitoring using sensors	Implement IoT sensors for real-time water quality monitoring with continuous data streaming	IoT system enables real-time monitoring of temperature and EC value.

Chapter 3

Objectives

- **Electrical Conductivity Management:** Monitoring and managing electrical conductivity ensures proper balance of dissolved ions and salinity, critical for maintaining water quality in small aquatic habitats.
- **Real-time Monitoring:** Providing immediate data on temperature and conductivity variations enables prompt adjustments, early detection of changes, and timely interventions to maintain ecosystem stability.
- **Efficient Resource Use:** Promoting efficient water management practices optimizes resource use, reduces water wastage and energy consumption, supports sustainability goals, and lowers operational costs.
- **Educational Value:** Water Quality Monitoring Systems offer invaluable educational opportunities, allowing stakeholders to learn hands-on about water quality management in small-scale settings.

Scope

- **Early Detection:** Continuous monitoring allows for early detection of changes in water quality, enabling proactive measures to prevent stress on aquatic life and disruptions to ecosystem dynamics.
- **Resource Management:** Real-time data on temperature and conductivity helps optimize water usage, minimize wastage, and adopt sustainable practices in small-scale aquatic environments. This conserves water resources, reduces operational costs, and promotes environmental sustainability.
- **Educational Applications:** The monitoring system is a valuable tool for hobbyists, educators, and researchers, offering hands-on learning about environmental monitoring and ecosystem management. It fosters awareness of water quality issues, encourages responsible stewardship, and supports scientific research and innovation in sustainable water management.

Problem Statement

- **Sensor Accuracy and Reliability:** Ensuring sensor accuracy and reliability in small-scale Water Quality Monitoring Systems (WQMS) is crucial for effective environmental management. Variations in sensor quality can compromise water quality assessments. High-quality, well-maintained, and regularly calibrated sensors are essential for precise measurement of parameters like temperature and electrical conductivity, ensuring trustworthy data for informed decision-making.

- **Calibration Requirements:** Regular sensor calibration is vital for accurate and consistent data collection in WQMS. Calibration aligns sensor measurements with known references and requires technical expertise and meticulous attention. Adhering to stringent calibration protocols maintains high measurement accuracy, even in dynamic small-scale environments with rapid fluctuations in temperature and conductivity.

- **Data Variability:** Small-scale environments often experience rapid changes in temperature and conductivity due to weather, biological activity, and human influences. Continuous monitoring and real-time data processing are necessary to capture these changes accurately. Effective management of data variability involves robust data processing algorithms and timely data transmission, enabling early anomaly detection, proactive decision-making, and prompt interventions to maintain ecosystem health.

Chapter 4

Block diagram

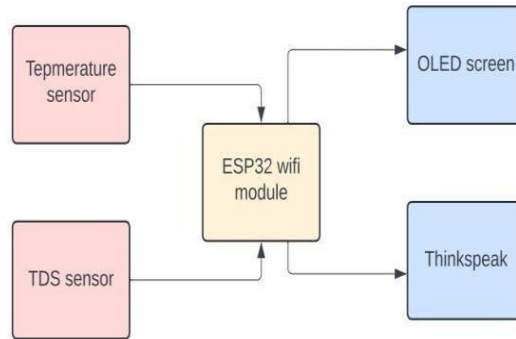


Fig. 4.1 : Block-diagram of the proposed methodology

- **Temperature Sensor:** Measures the water temperature and sends the data to the ESP32Wi-Fi module.
- **TDS Sensor:** Measures the Total Dissolved Solids (TDS) in the water and sends the data to the ESP32 Wi-Fi module.
- **ESP32 Wi-Fi Module:** Acts as the central processing unit. It collects data from the sensors, processes it, and sends it to both the OLED screen for local display and Thinkspeak for remote monitoring.
- **OLED Screen:** Displays real-time water quality data locally for immediate viewing.
- **ThinkSpeak:** An IoT analytics platform where the data is sent for remote monitoring and further analysis. It allows users to access and visualize water quality data from anywhere.

Circuit Diagram

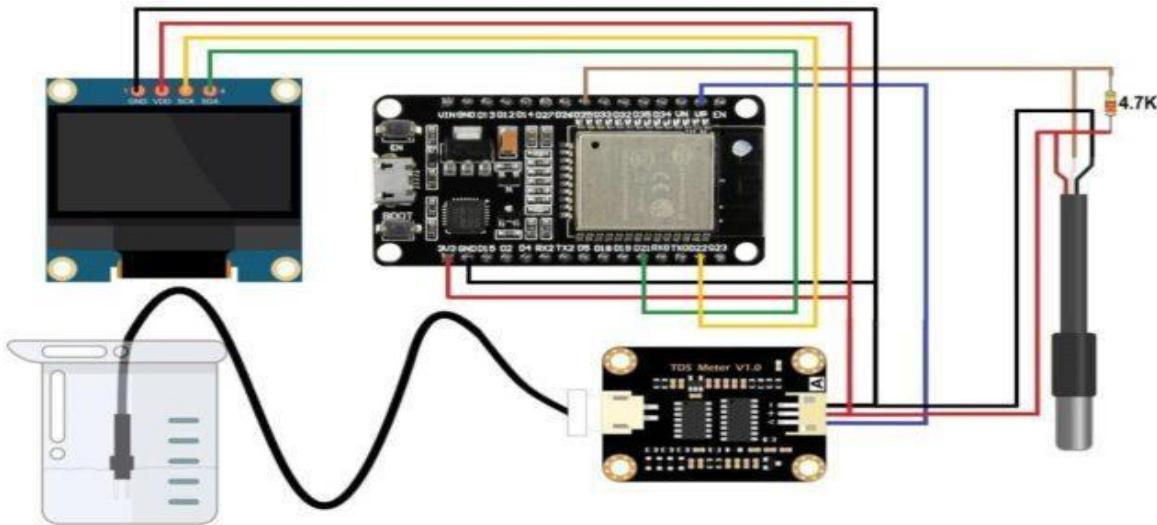


Fig. 4.2 : Circuit Diagram

Working Principle

This water quality monitoring system measures the temperature of the water and also the electrical conductivity of the water.

After uploading the code to the ESP32 Board, the ESP32 will try connecting to the WiFi Network. Once it connects to the WiFi Network, it will start reading the EC & Temperature values from the Sensor. You can open the Serial Monitor to read those values.

The temperature sensor will show the room temperature. When the TDS Sensor is dry and not exposed to air, it will show an EC value as zero. You can view these data on OLED Screen.

When the temperature Sensor & TDS Sensor is dipped in water, the OLED will display the EC & Temperature value of the water.

Adding more salts to water increases the electrical conductivity of water & will show a massive rise in water EC value.

You can also monitor the same data on the Thingspeak dashboard. To view the data, go to the Thingspeak private view & you will see the EC & Temperature data getting uploaded after the interval of every 15 seconds.

Algorithm

Algorithm or flowchart of the mini-project work should be explained in greater detail. In this chapter, how the flow-chart or the data flow diagram or the algorithm that has been developed & used has to be presented with discussion on each and every blocks-sets.

This Arduino sketch is designed for the ESP32 microcontroller and integrates various sensors and modules to monitor and display environmental parameters, specifically temperature and electrical conductivity (EC). Here's a breakdown of what the code does and how it is structured:

Libraries Used:

- *Wire.h*: I2C communication library.
- *EEPROM.h*: Library for EEPROM read/write.
- *WiFi.h*: Library to connect to WiFi networks.
- *OneWire.h*: Communicates with devices using the OneWire protocol.

- DallasTemperature.h: Reads temperatures from OneWire temperature sensors like DS18B20.
- Adafruit_ADS1015.h: Interface for ADS1015/ADS1115 ADC.
- DFRobot_ESP_EC.h: Measures Electrical Conductivity (EC).
- Adafruit_GFX.h: Core graphics library.
- Adafruit_SSD1306.h: Driver for SSD1306 OLED displays.

Hardware and Connections:

- OLED Display: 128x64 pixels, I2C interface connected via Wire library.
- DS18B20 Temperature Sensor: Connected to GPIO 14 using OneWire protocol.
- ADS1115 ADC: Reads analog values (though in the code analogRead is used, which is specific to ESP32's internal ADC).
- EC Sensor: Reads electrical conductivity, connected through the DFRobot library.

Key Components and Definitions:

- *Display Initialization*: Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1); initializes the OLED display with specified dimensions.
- *OneWire and DallasTemperature*: Handles temperature readings from the DS18B20 sensor.
- *WiFi Connection*: Connects to a specified WiFi network for data transmission.
- *ThingSpeak*: API integration for data logging to ThingSpeak.

Code Workflow:

- *Setup Function (setup)*:
 - Initializes serial communication for debugging.
 - Sets up EEPROM for storing calibration data.
 - Initializes temperature sensors.
 - Initializes the OLED display.
 - Connects to the specified WiFi network.
 - Prints connection status.

- Main Loop (loop)*:
- Reads the voltage from an analog pin (A0 mapped to GPIO 36).
- Requests and reads the temperature from the DS18B20 sensor.
- Computes the EC value using the temperature-compensated voltage.
- Displays temperature and EC on the OLED screen.
- Conducts EC sensor calibration.
- Sends data to ThingSpeak if a connection is available.

Flowchart

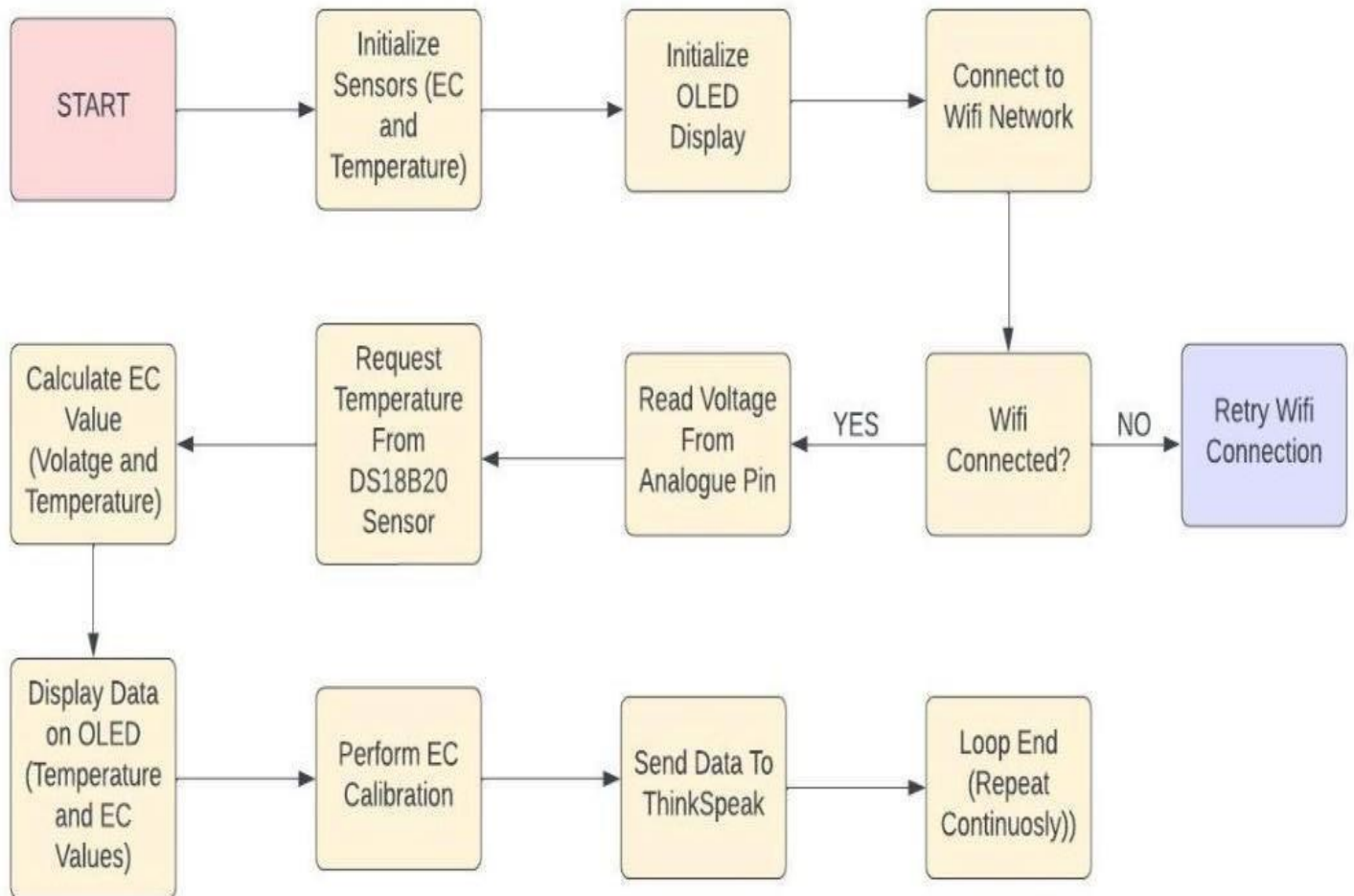


Fig. 4.4 : Flow-chart of the methodology used

Explanation:

1. Start: The system powers on and begins initialization.

2. Initialization Phase:

- Serial communication starts to enable debugging and monitoring.
- EEPROM is initialized to manage calibration data.
- EC and temperature sensors are initialized.
- OLED display is set up.
- WiFi connection is established.

3. Main Loop:

- The system reads the voltage from the analog pin connected to the EC sensor.
- It requests temperature data from the DS18B20 sensor.
- The EC value is calculated based on the voltage and temperature.
- Temperature and EC values are displayed on the OLED.
- EC sensor calibration is performed if necessary.
- The collected data is sent to ThingSpeak for logging and monitoring.

4. Loop End:

- The process repeats indefinitely, continuously monitoring water quality parameters.

This flowchart effectively illustrates the sequential flow of operations in a small-scale Water Quality Monitoring System.

Chapter-5

Hardware :

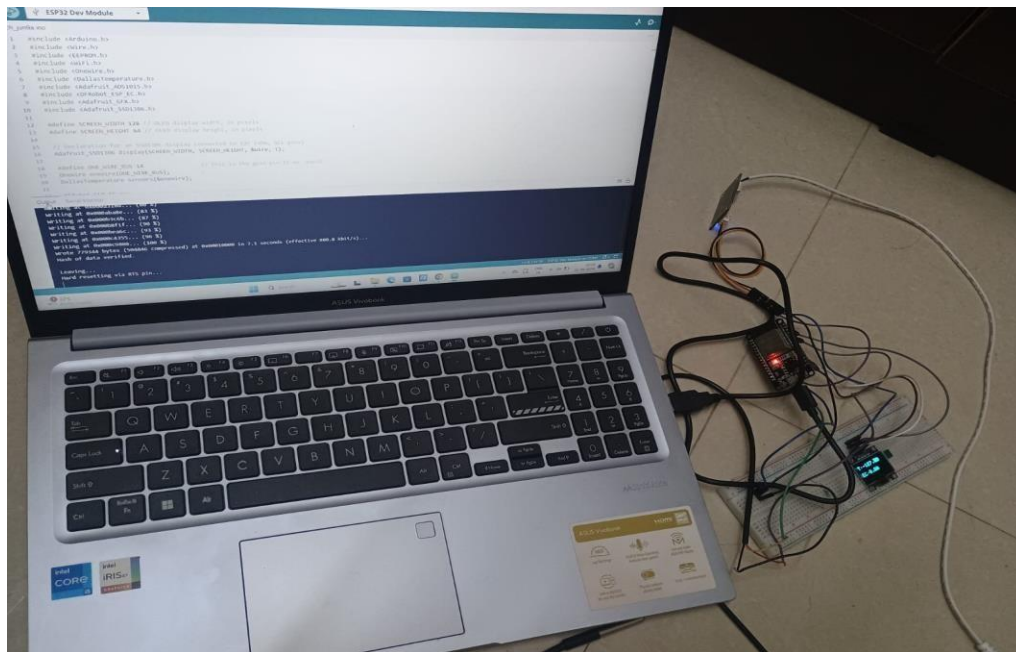
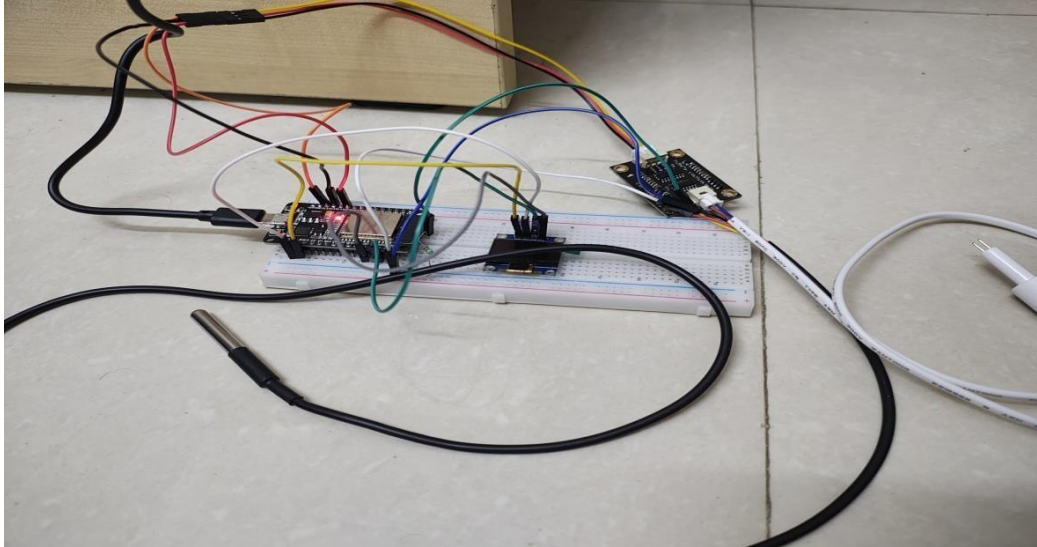
- **ESP32 Board:** The ESP32 Board is a highly versatile microcontroller featuring integrated Wi-Fi and Bluetooth capabilities, making it an ideal choice for various IoT (Internet of Things) applications. It serves as the central processing unit in this setup, capable of interfacing with other components to gather, process, and transmit data wirelessly.
- **Analog TDS Sensor:** An Analog TDS Sensor is employed to measure the total dissolved solids (TDS) in water. This sensor provides an analog signal that correlates with the TDS level of the water being tested, offering valuable insights into water quality.
- **DS18B20 Temperature Sensor:** For precise temperature monitoring, the DS18B20 Temperature Sensor is utilized. This digital sensor communicates over a 1-Wire bus, enabling accurate temperature measurements with minimal wiring complexity.
- **Resistor 4.7K:** A Resistor 4.7K is essential in this setup as a pull-up resistor for the DS18B20 sensor, ensuring stable and reliable communication on the 1-Wire bus.
- **Connecting Wires:** Connecting Wires play a crucial role in establishing electrical connections between the various components of the system. They facilitate seamless data and power transmission throughout the circuit, ensuring proper functionality.
- **Breadboard:** To facilitate rapid prototyping and testing, a Breadboard is employed. This prototyping board allows components to be interconnected without soldering, making it easier to assemble and modify circuits during development.
- **OLED Display:** An OLED Display serves as a local interface for the system, providing real-time visualization of data collected from the sensors. It enables users to monitor TDS levels, temperature readings, and other pertinent information directly on the display, enhancing usability and accessibility.

Software :

- Arduino IDE: For programming the ESP32 microcontroller.
- ESP32 Libraries: Specific libraries for handling Wi-Fi connectivity, sensor integration, and display output.
- Dashboard Software: Thingspeak-for monitoring the parameters(temperature & conductivity).
- Microprocessor language : c++

Chapter-6

Photographs of the model



Chapter-7

Results and Discussions

This water quality monitoring system measures water temperature and electrical conductivity (EC). After uploading the code to the ESP32 Board, it connects to a WiFi network. Upon connection, it begins reading temperature and EC values from the sensors. The temperature sensor initially shows room temperature, while the EC sensor shows zero when dry. These values are displayed on an OLED screen. When the sensors are immersed in water, the OLED screen updates to show the water's EC and temperature values. Adding salt to the water increases its electrical conductivity, causing a significant rise in the EC value displayed. Additionally, data is uploaded to the Thingspeak dashboard every 15 seconds. To view this data, you can access the Thingspeak private view, where you will see the updated EC and temperature readings. This continuous monitoring and real-time data transmission allow for efficient tracking and management of water quality parameters.

Chapter-8

Applications

1. Environmental Monitoring:

- **Temperature:** Tracks thermal pollution from industrial discharges and climate change, impacting aquatic ecosystems and habitat suitability.
- **Electrical Conductivity:** Detects pollution and natural changes in water composition, highlighting contamination risks from agriculture, mining, or urban activities.

2. Drinking Water Quality Assurance:

- **Temperature:** Ensures compliance with water temperature regulations during distribution to prevent microbial growth and maintain quality.
- **Electrical Conductivity:** Monitors total dissolved solids (TDS), assessing water purity and contamination sources.

3. Industrial Process Control:

- **Temperature:** Critical for precise control in manufacturing, cooling, and chemical processes involving water.
- **Electrical Conductivity:** Monitors industrial discharge to comply with environmental regulations, minimizing impact on ecosystems.

Advantages

Real-Time Monitoring:

- **Temperature:** Provides immediate insights into water temperature variations critical for aquatic ecosystems.
- **Electrical Conductivity:** Enables real-time detection of water composition changes, indicating pollution or natural fluctuations.

Cost-Effectiveness:

- More affordable and easier to deploy than large-scale networks.
- Scalable for targeted monitoring in localized water bodies.

Regulatory Compliance:

- Facilitates continuous monitoring for environmental assessments.
- Ensures adherence to water quality standards and mitigates environmental risks..

Research and Education:

- Provides data for scientific research on hydrology and ecosystem health.
- Enhances educational opportunities with practical environmental monitoring activities.

Outcomes

- **Early Detection:** Monitoring detects deviations early, signaling equipment issues or environmental changes, allowing prompt corrective actions to safeguard water quality and aquatic life.
- **Optimized Conditions:** Continuous monitoring ensures optimal temperature ranges for aquatic species and appropriate ion concentrations crucial for their health.
- **Efficient Resource Use:** Helps optimize energy and nutrient resources, minimizing wastage and promoting sustainability in water and energy consumption.
- **Educational Tools:** Serve as hands-on educational tools for learning water quality management principles and fostering environmental stewardship among users.
- **Cost-Effective and Scalable:** Affordable and scalable solutions suitable for various settings, from homes to educational and research institutions, enhancing ecosystem health locally.

Limitations

- Limited scope, as they only monitor temperature and electrical conductivity, missing other crucial water quality factors.
- Localized monitoring may not capture broader environmental trends.
- Sensor sensitivity issues can affect data accuracy in dynamic water conditions.
- Resource constraints limit deployment in larger areas or multiple sites.
- Challenges in scalability and integration with existing infrastructure.
- Expertise needed for interpreting and applying data effectively for management decisions.

Chapter-9

Conclusions and Future Work

Conclusions and future work for Water Quality Monitoring Systems (WQMS) focusing on temperature and conductivity parameters underscore several key points:

- **Effective Monitoring Capability:** Despite their limitations, small-scale WQMS provide valuable real-time data on temperature and conductivity, which are crucial for understanding immediate water quality conditions in localized environments.
- **Enhanced Data Integration:** Future developments could focus on integrating additional sensors to broaden the scope of monitored parameters, providing more comprehensive insights into water quality dynamics.
- **Technological Advancements:** Continued advancements in sensor technology can improve accuracy, sensitivity, and robustness, enhancing the reliability of data collected by small-scale WQMS.
- **Data Analysis and Interpretation:** There is a need for sophisticated algorithms and data analytics techniques to effectively process and interpret the collected data. This would enable better decision-making for managing water resources and mitigating environmental risks.
- **Scalability and Accessibility:** Addressing challenges related to scalability and affordability will be crucial for expanding the deployment of small-scale WQMS across diverse geographic and environmental settings.
- **Interdisciplinary Collaboration:** Future research could benefit from interdisciplinary collaboration between scientists, engineers, policymakers, and stakeholders to address technical challenges and translate data into actionable insights for sustainable water management.

In essence, while small-scale WQMS focusing on temperature and conductivity parameters provide a foundation for monitoring water quality, ongoing research and development efforts are essential to enhance their capabilities, broaden their applicability, and maximize their impact on environmental stewardship and resource conservation.

References

Journal papers –

- Singh, A., & Jain, S. (2017). "Internet of Things (IoT) for Smart Water Quality Monitoring:A Comprehensive Review." Journal of Ambient Intelligence and Humanized Computing, 8(2)
- Jain, A, et al. (2019). "An IoT-Based Real-Time Water Quality Monitoring System." IEEE Access,7

Websites –

- [1] [IoT Based Drinking Water Quality Monitoring with ESP32 \(how2electronics.com\)](http://how2electronics.com)
- [2] [Journal of Microelectronics and Solid State Devices \(stmjournals.in\)](http://stmjournals.in)

For Books –

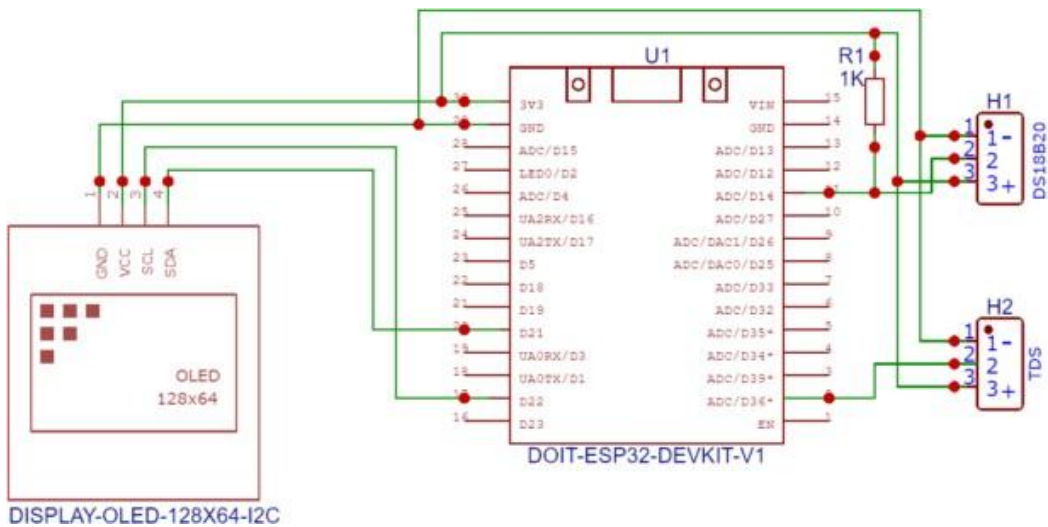
- [1] "Design of Water Quality Monitoring Systems" by Robert C. Ward and Jack C. Loftis

Overview: This book focuses on the design aspects of water quality monitoring systems. It provides a structured approach to developing effective monitoring networks and systems tailored to specific objectives.

- [2] "Water Quality Monitoring: A Practical Guide to the Design and Implementation of Freshwater Quality Studies and Monitoring Programmes" by Jamie Bartram and Richard Ballance

Overview: This book provides a thorough guide to designing and implementing water quality monitoring programs. It includes practical advice on selecting sampling sites, choosing appropriate methods, and interpreting results.

Appendix



Code:

```
#include <Arduino.h>
#include <Wire.h>
#include <EEPROM.h>
#include <WiFi.h>
#include <OneWire.h>
#include <DallasTemperature.h>
#include <Adafruit_ADS1015.h>
#include <DFRobot_ESP_EC.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128 // OLED display width, in pixels
#define SCREEN_HEIGHT 64 // OLED display height, in pixels

// Declaration for an SSD1306 display connected to I2C (SDA, SCL pins)
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

#define ONE_WIRE_BUS 14 // this is the gpio pin 13 on esp32.
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);

DFRobot_ESP_EC ec;
```

```
Adafruit_ADS1115 ads;

float voltage, ecValue, temperature = 25;

String apiKey = "*****"; // Enter your Write API key from ThingSpeak

const char *ssid = "XD"; // replace with your wifi ssid and wpa2 key
const char *pass = "ayushhhh";
const char* server = "api.thingspeak.com";

WiFiClient client;

void setup()
{
  Serial.begin(115200);
  EEPROM.begin(32); //needed EEPROM.begin to store calibration k in eeprom
  ec.begin();
  sensors.begin();
  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) { // Address 0x3D for 128x64
    Serial.println(F("SSD1306 allocation failed"));
    for (;;)
      ;
  }
  delay(2000);
  display.clearDisplay();

  Serial.println("Connecting to ");
  Serial.println(ssid);

  WiFi.begin(ssid, pass);

  while (WiFi.status() != WL_CONNECTED)
  {
    delay(500);
    Serial.print(".");
  }
  Serial.println("");
  Serial.println("WiFi connected");
}

void loop()
{
  voltage = analogRead(A0); // A0 is the gpio 36
  sensors.requestTemperatures();
  temperature = sensors.getTempCByIndex(0); // read your temperature sensor to execute
  temperature compensation
  ecValue = ec.readEC(voltage, temperature); // convert voltage to EC with temperature
  compensation

  Serial.print("Temperature:");
```

```
Serial.print(temperature, 2);
Serial.println("°C");

Serial.print("EC:");
Serial.println(ecValue, 2);

display.setTextSize(2);
display.setTextColor(WHITE);

display.setCursor(0, 10);
display.print("T:");
display.print(temperature, 2);
display.drawCircle(85, 10, 2, WHITE); // put degree symbol ( ° )
display.setCursor(90, 10);
display.print("C");

display.setCursor(0, 40);
display.print("EC:");
display.print(ecValue, 2);
display.display();
delay(1500);
display.clearDisplay();

ec.calibration(voltage, temperature); // calibration process by Serail CMD

if (client.connect(server, 80)) // "184.106.153.149" or api.thingspeak.com
{
String postStr = apiKey;
postStr += "&field1=";
postStr += String(temperature, 2);
postStr += "&field2=";
postStr += String(ecValue, 2);
postStr += "\r\n\r\n";
delay(500);

client.print("POST /update HTTP/1.1\n");
client.print("Host: api.thingspeak.com\n");
client.print("Connection: close\n");
client.print("X-THINGSPEAKAPIKEY: " + apiKey + "\n");
client.print("Content-Type: application/x-www-form-urlencoded\n");
client.print("Content-Length: ");
client.print(postStr.length());
client.print("\n\n");
client.print(postStr);
delay(500);
}
client.stop();
}
```